

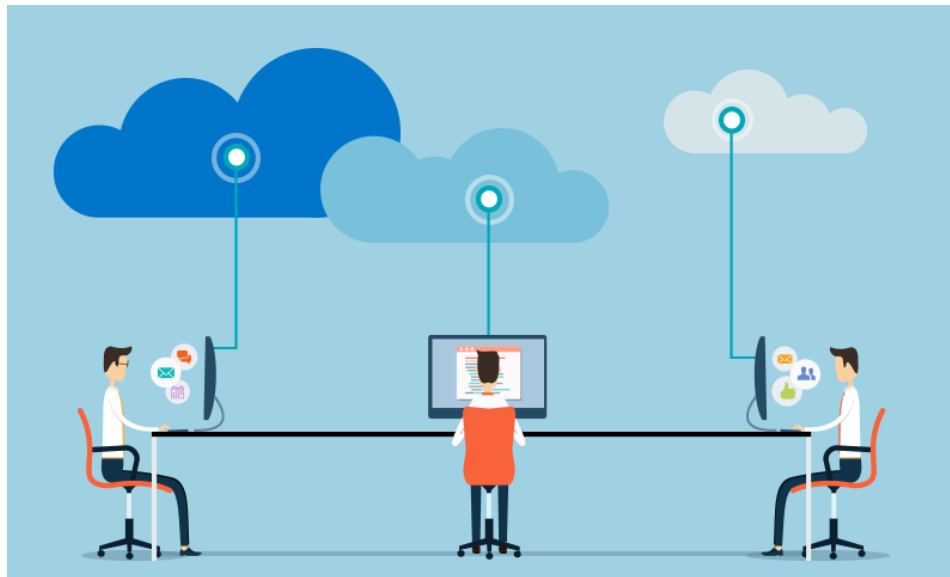


TECNOLÓGICO
NACIONAL DE MÉXICO



**Tecnológico Nacional de México
Instituto Tecnológico de Saltillo**

Departamento de Ingeniería en Sistemas Computacionales



Materia: Computo en la Nube

Profesor: Salazar Del Bosque Miguel

(7 – 8 hrs)

“5.9 Despliegue de aplicación CRUD en la nube”

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No. De Control: 21051534

```

PS C:\WINDOWS\system32> & "C:\Windows\System32\OpenSSH\scp.exe" -i "C:\Users\jesus\Documents\10 semestre
\Computo en la Nube\MV-CN_key.pem" -r "C:\Users\jesus\Documents\10 semestre\Computo en la Nube\USERS" az
ureuser@4.154.236.115:/home/azureuser/
.DS_Store                                100% 6148    54.1KB/s   00:00
.DS_Store                                100% 6148    47.7KB/s   00:00
controlador.php                          100% 899     7.8KB/s    00:00
ControladorUsuario.php                   100% 1564    16.1KB/s    00:00
.DS_Store                                100% 6148    53.6KB/s    00:00
Captura de pantalla 2025-09-21 a la(s) 8.01.470Ç»a.m..png 100% 498KB 576.1KB/s   00:00
escudoITS.png                            100% 20KB   157.2KB/s   00:00
tecnm.jpg                                100% 26KB   221.5KB/s   00:00
tecnm.png                                100% 25KB   173.1KB/s   00:00
user.png                                  100% 6196    63.0KB/s    00:00
index.php                                100% 253     2.6KB/s     00:00
.DS_Store                                100% 6148    50.0KB/s    00:00
Captura de pantalla 2025-09-29 a la(s) 1.10.510Ç»p.m..png 100% 81KB  285.6KB/s   00:00
Database.php                             100% 524     5.4KB/s     00:00
modeloUsuario.php                       100% 2066    18.2KB/s     00:00
.DS_Store                                100% 6148    54.6KB/s     00:00
.DS_Store                                100% 6148    48.8KB/s     00:00
all.min.css                              100% 57KB   238.4KB/s    00:00
bootsnip.css                             100% 5642    49.2KB/s     00:00
bootstrap-3.3.5.min.css                  100% 118KB  342.0KB/s     00:00
bootstrap-3.3.5.min1.css                 100% 118KB  371.0KB/s     00:00
bootstrap-theme.min.css                  100% 19KB   203.1KB/s     00:00
jquery-ui.min.css                        100% 29KB   241.5KB/s     00:00
jQuery.dataTables.min.css                100% 14KB   122.3KB/s     00:00
sb-admin-2.css                           100% 199KB  512.0KB/s     00:00
sb-admin-2.min.css                       100% 161KB  460.2KB/s     00:00
style.css                                100% 272     2.3KB/s     00:00
datatable.js                              100% 710     6.1KB/s     00:00
dataTables.bootstrap.css                 100% 4823    36.8KB/s     00:00
dataTables.bootstrap.js                  100% 4559    46.9KB/s     00:00
dataTables.bootstrap.min.js              100% 1966    15.0KB/s     00:00
autoFill.bootstrap.css                   100% 1895    19.1KB/s     00:00
autoFill.bootstrap.min.css               100% 1615    16.3KB/s     00:00
autoFill.bootstrap4.css                  100% 1895    16.5KB/s     00:00
autoFill.bootstrap4.min.css              100% 1615    15.3KB/s     00:00
autoFill.dataTables.css                  100% 2192    14.1KB/s     00:00
autoFill.dataTables.min.css              100% 1872    14.4KB/s     00:00

```

Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.17.0-1011-azure x86_64)

* Documentation: <https://help.ubuntu.com>
* Management: <https://landscape.canonical.com>
* Support: <https://ubuntu.com/pro>

System information as of Fri May 1 00:17:19 UTC 2026

System load:	0.01	Processes:	145
Usage of /:	8.8% of 28.02GB	Users logged in:	1
Memory usage:	9%	IPv4 address for eth0:	10.0.0.4
Swap usage:	0%		

* Strictly confined Kubernetes makes edge and IoT secure. Learn how MicroK8s just raised the bar for easy, resilient and secure K8s cluster deployment.

<https://ubuntu.com/engage/secure-kubernetes-at-the-edge>

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See <https://ubuntu.com/esm> or run: `sudo pro status`

Last login: Thu Apr 30 23:56:23 2026 from 4.151.247.81

azureuser@MV-CN:~\$ cd /home/azureuser/

azureuser@MV-CN:~\$ ls

USERS

azureuser@MV-CN:~\$ █

Microsoft Azure

Buscar recursos, servicios y documentos

Cambiar a PowerShell

Reiniciar

Administrar archivos

Nueva sesión

Editor

Vista previa

```
azureuser@MV-CN:~$ cd /home/azureuser/
azureuser@MV-CN:~$ ls
USERS
azureuser@MV-CN:~$ sudo rm -rf /var/www/html/*
azureuser@MV-CN:~$ sudo mv USERS/* /var/www/html/
azureuser@MV-CN:~$
azureuser@MV-CN:~$ sudo mysql -u root
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 10
Server version: 8.0.45-0ubuntu0.24.04.1 (Ubuntu)

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owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> SHOW DATABASES;
+-----+
| Database |
+-----+
| information_schema |
| mysql |
| performance_schema |
| sys |
+-----+
4 rows in set (0.00 sec)

mysql>
```

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Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

```
mysql> select * from usuarios
```

```
->
```

```
-> ;
```

ERROR 1046 (3D000): No database selected

```
mysql> use mysql;
```

Reading table information for completion of table and column names

You can turn off this feature to get a quicker startup with -A

Database changed

```
mysql> select * from usuarios;
```

ERROR 1146 (42S02): Table 'mysql.usuarios' doesn't exist

```
mysql> use db_usuarios;
```

Reading table information for completion of table and column names

You can turn off this feature to get a quicker startup with -A

Database changed

```
mysql> select * from usuarios;
```

id	nombre	correo
1	Jesus Vazquez	121051534@saltillo.tecnm.mx

1 row in set (0.00 sec)

```
mysql>
```

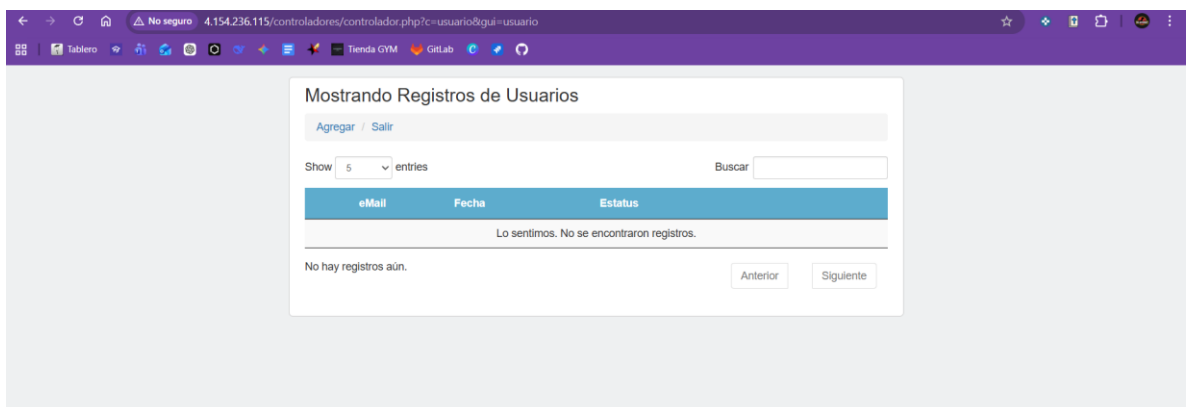
```
Microsoft Azure
Buscar recursos, servicios y documentos (G+/)

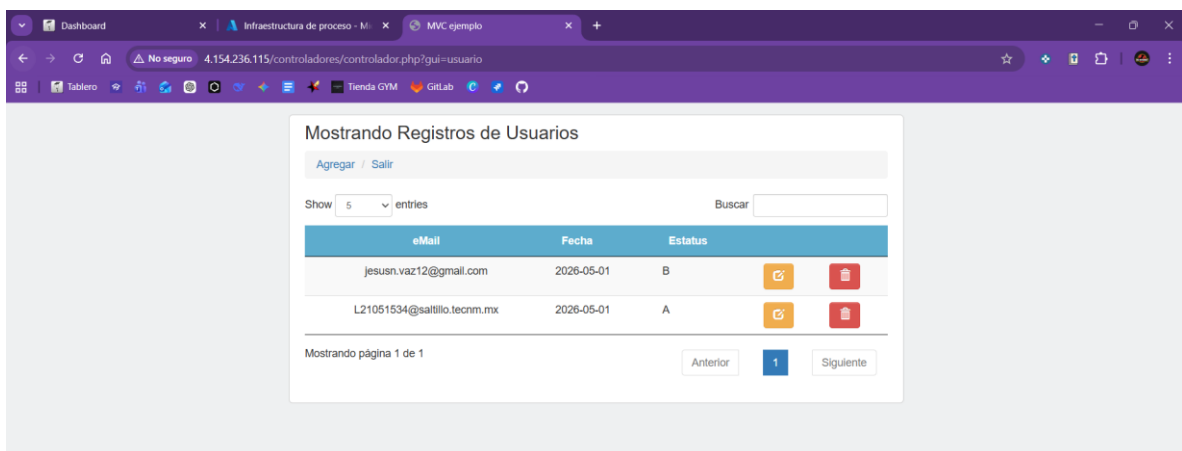
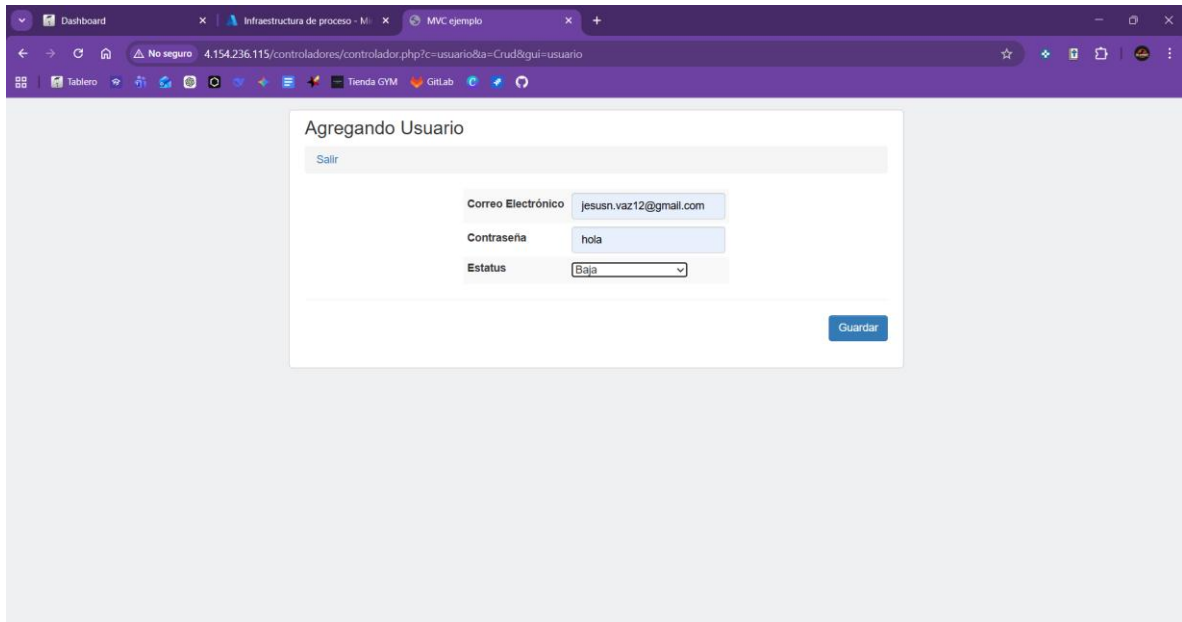
Cambiar a PowerShell Reiniciar Administrar archivos Nueva sesión Editor Vista previa web Configuración

/var/www/html/usuarios/vista:
headerHTML.php usuario

/var/www/html/usuarios/vista/usuario:
shwUsuario.php updUsuario.php
azureuser@MV-CN:~$ cd /var/www/html/usuarios/
azureuser@MV-CN:/var/www/html/usuarios$ sudo mv * /var/www/html/
azureuser@MV-CN:/var/www/html/usuarios$ sudo rm -rf /var/www/html/__MACOSX
sudo rm -rf /var/www/html/usuarios
azureuser@MV-CN:/var/www/html/usuarios$ sudo nano /var/www/html/modelo/Database.php
azureuser@MV-CN:/var/www/html/usuarios$ cat /var/www/html/modelo/Database.php
<?php

class Database
{
    public static function StartUp()
    {
        //$pdo = new PDO('mysql:host=172.16.10.156;port=3306;dbname=cNube;charset=utf8', 'msb', 'admin.');
```





Códigos:

```
azureuser@MV-CN:/var/www/html/modelo$ cat modeloUsuario.php
```

```
<?php
```

```
class modeloUsuario{
```

```
    private $pdo;
```

```
    public $id;
```

```
    public $eMail;
```

```
    public $pwd_;
```

```
    public $fecha;
```

```
    public $estatus;
```

```
    public function __CONSTRUCT()
```

```
    {
```

```
        try {
```

```

        $this->pdo = Database::StartUp();
    } catch(Exception $e) {
        die($e->getMessage());
    }
}

```

```

public function Listar()
{
    try {
        $sql = "SELECT * FROM usuario ORDER BY id DESC";
        $stm = $this->pdo->prepare($sql);
        $stm->execute();
        return $stm->fetchAll(PDO::FETCH_OBJ);
    } catch(Exception $e) {
        die($e->getMessage());
    }
}

```

```

public function Obtener($id)
{
    try {
        $stm = $this->pdo->prepare("SELECT * FROM usuario WHERE id = ?");
        $stm->execute(array($id));
        return $stm->fetch(PDO::FETCH_OBJ);
    } catch (Exception $e) {
        die($e->getMessage());
    }
}

```

```

public function Eliminar($id)
{
    try {
        $stm = $this->pdo->prepare("DELETE FROM usuario WHERE id = ?");
        $stm->execute(array($id));
    } catch (Exception $e) {
        die($e->getMessage());
    }
}

```

```

public function Actualizar($data)
{
    try {
        $sql = "UPDATE usuario SET eMail = ?, estatus = ? WHERE id = ?";
        $this->pdo->prepare($sql)->execute(
            array($data->eMail, $data->estatus, $data->id)
        );
    } catch (EXCEPTION $e) {

```



```

        die($e->getMessage());
    }
}

public function Registrar($data)
{
    // Iniciamos sesión si no está iniciada para manejar el errMsg
    if (session_status() == PHP_SESSION_NONE) { session_start(); }
    $_SESSION['errMsg'] = 0;

    try {
        $sql = "INSERT INTO usuario (eMail, pwd_, fecha, estatus) VALUES (?, ?, ?,
    ?)";
        $this->pdo->prepare($sql)->execute(
            array($data->eMail, $data->pwd_, $data->fecha, $data->estatus)
        );
    } catch (EXCEPTION $e) {
        $_SESSION['errMsg'] = 1;
        error_log($e->getMessage());
    }
}
}
?>

```

azureuser@MV-CN:/var/www/html/modelo\$

azureuser@MV-CN:/var/www/html/modelo\$ cat Database.php

```

<?php
class Database
{
    public static function StartUp()
    {
        // Usamos el nombre de la base de datos que definiste: db_usuarios
        $dsn =
"mysql:host=localhost;port=3306;dbname=db_usuarios;charset=utf8mb4";
        $pdo = new PDO($dsn, "admin", "1234", [
            PDO::ATTR_ERRMODE => PDO::ERRMODE_EXCEPTION,
            PDO::ATTR_DEFAULT_FETCH_MODE => PDO::FETCH_ASSOC,
        ]);

        return $pdo;
    }
}
?>

```

azureuser@MV-CN:/var/www/html/modelo\$

**azureuser@MV-CN:/var/www/html/controladores\$ cat
ControladorUsuario.php**

```

<?php
require_once('../modelo/modeloUsuario.php');

class ControladorUsuario{

    private $modelo;

    public function __CONSTRUCT(){
        $this->modelo = new modeloUsuario();
    }

    public function Index(){
        require_once '../vista/headerHTML.php';
        require_once '../vista/usuario/shwUsuario.php';
    }

    public function Crud(){
        $entidadUsuario = new modeloUsuario();
        if(isset($_REQUEST['id'])){
            $entidadUsuario = $this->modelo->Obtener($_REQUEST['id']);
        }
        require_once '../vista/headerHTML.php';
        require_once '../vista/usuario/updUsuario.php';
    }

    public function Guardar(){
        if (session_status() == PHP_SESSION_NONE) { session_start(); }
        $entidadUsuario = new modeloUsuario();

        $entidadUsuario->id      = $_REQUEST['txtId'];
        $entidadUsuario->eMail   = $_REQUEST['txtEMail'];
        $entidadUsuario->pwd_    = $_REQUEST['txtPwd'];
        $entidadUsuario->fecha   = date('Y-m-d');
        $entidadUsuario->estatus = $_REQUEST['selEstatus'];

        if($entidadUsuario->id > 0){
            $this->modelo->Actualizar($entidadUsuario);
            header('Location: ./controlador.php?gui=usuario');
        }
        else {
            $this->modelo->Registrar($entidadUsuario);

            // Si no hay error, regresamos a la lista de usuarios
            if($_SESSION['errMsg'] == 0) {
                header('Location: ./controlador.php?gui=usuario');
            } else {
                // Si hay error, volvemos a mostrar el formulario con el error
            }
        }
    }
}

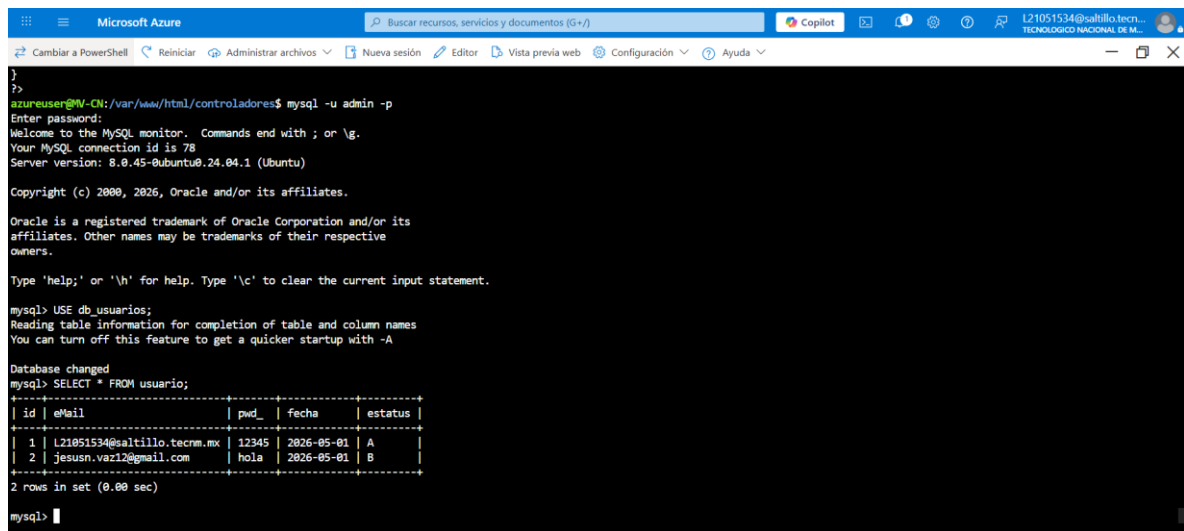
```

```

        require_once '../vista/headerHTML.php';
        require_once '../vista/usuario/updUsuario.php';
    }
}
}

public function Eliminar(){
    $this->modelo->Eliminar($_REQUEST['id']);
    header('Location: ./controlador.php?gui=usuario');
}
}
?>
azureuser@MV-CN:/var/www/html/controladores$

```



The screenshot shows a terminal window with the following content:

```

azureuser@MV-CN:/var/www/html/controladores$ mysql -u admin -p
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 78
Server version: 8.0.45-0ubuntu0.24.04.1 (Ubuntu)

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owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> USE db_usuarios;
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Database changed
mysql> SELECT * FROM usuario;
+----+-----+-----+-----+-----+
| id | eMail | pwd_ | fecha | estatus |
+----+-----+-----+-----+-----+
| 1  | L21051534@saltillo.tecnm.mx | 12345 | 2026-05-01 | A |
| 2  | Jesusn.var12@gmail.com | hola | 2026-05-01 | B |
+----+-----+-----+-----+-----+
2 rows in set (0.00 sec)

mysql>

```